

Mahi Patel

m19patel@torontomu.ca | 365-999-1610 | Markham, ON | LinkedIn: <https://www.linkedin.com/in/mahiptl/> | GitHub: [mahip16](#)

QUALIFICATIONS

- 2+ years building full-stack and backend applications in Java, Python, and JavaScript.
- Shipped 2 production-ready projects from scratch: a cross-platform mobile app with real-time cloud sync and a live ML web app with automated retraining deployed on Vercel.
- Hands-on experience with machine learning workflows, including feature engineering, model training, time-aware validation, and Monte Carlo simulation.
- Strong foundation in object-oriented programming, data structures, REST API integration, and debugging.
- Experienced with SQL and NoSQL databases (Firebase Firestore, Realtime DB) and cloud/serverless platforms, including Firebase and AWS fundamentals.
- Eligible for 4-, 8-, or 12-month co-op work term.

EDUCATION

Toronto Metropolitan University

Sep. 2024 - Expected June. 2029

Bachelor of Science in Computer Science, Co-operative Education,

Toronto, ON

Technical Skills:

- **Languages:** Java, Python, JavaScript, HTML, CSS, C
- **Frameworks and Tools:** React, React Native, Expo, Streamlit, Git, GitHub, Swing
- **Databases:** SQL (fundamentals), Firebase Firestore, Firebase Realtime Database
- **Machine Learning:** XGBoost, scikit-learn, Pandas, NumPy, Plotly, Monte Carlo Simulation, Feature Engineering
- **Concepts:** OOP, Data Structures and Algorithms, REST APIs, CRUD Operations, CI/CD, Unit Testing
- **Practices:** Agile, SDLC, Version Control, GitHub Actions, Cloud Deployment (Vercel, Firebase)

PROJECTS

F1 Race Win Predictor | Python, XGBoost, Streamlit, Pandas | Machine Learning

Mar. 2026

Live Demo: <https://f1-race-predictor-taupe.vercel.app/>

- Achieved 0.088 log-loss on a fully held-out 2025 season by training an XGBoost classifier on 15 years of F1 data across 6,900+ race entries, outperforming a naive baseline by roughly 3x.
- Prevented data leakage across all rolling features by applying shift(1) to driver form, circuit win rate, and team performance metrics, ensuring the model never saw future race data during training.
- Produced calibrated win probability distributions by running 10,000 Monte Carlo simulations per race, modelling DNF rates, safety car probabilities, and on-track performance variation.
- Eliminated manual deployment by building a GitHub Actions pipeline that fetches new race results, retrains the model, and redeploys to Vercel automatically after each Grand Prix.
- Extended prediction capability to the live 2026 season by building a qualifying-result input system that generates real-time win probabilities for upcoming races.

Supplement Verification & Tracking App | React Native, Firebase | Full Stack

Sep. 2025 - Dec. 2025

- Delivered real-time supplement safety insights to users by integrating the Open Food Facts API with a custom ingredient analysis layer that flags allergens, caffeine content, and risky additives, surfacing a 0-10 safety score on every scan.
- Secured all user data and enabled cross-device access by implementing role-based Firebase Auth with email/password login, guest mode, and real-time Firestore sync across scan history, favourites, and profiles.
- Supported accurate scanning across diverse product types by building barcode detection for UPC, EAN, Code128, and QR formats with low-light flash toggling using the Expo Camera API.
- Improved user retention and daily engagement by designing a personalized health dashboard with scan statistics, daily supplement tips, search, and pull-to-refresh history filtering.

